PRE-ALGEBRA (7TH) · UNIT 6: RATIOS, PROPORTIONS, AND PERCENTAGES

Sneaker Store Manager

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students manage a sneaker store. They compare supplier unit rates, use proportions for orders, use a scale drawing and similar figures to lay out the store, run a raffle with probability, and set prices with markup, discount, and percent change.

Every percent in the unit has a real effect on profit, which motivates careful reasoning.

Standards Addressed

- 7.RP.A.1–2 Unit rates and proportional relationships
- 7.RP.A.3 Percent problems: markup, discount, percent change
- 7.G.A.1 Scale drawings
- 7.SP.C.5–8 Probability

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a sneaker ad. Part 1 suppliers and unit rates.
Day 2	Part 2 store layout and scale.
Day 3	Part 3 pricing, raffle, and the manager report.

Materials

- Student handout and Supplier Cards
- Rulers and grid paper
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Unit prices, one-step proportions, a scale drawing, single markup and discount, and simple probability.
CP1	Ranking suppliers, percent saved, similar figures, percent change, and expected wins in a raffle.
Honors	Pack constraints with a budget, area scale factors, compound discounts, probability without replacement, and a pricing inequality.

Watch For (Common Misconceptions)

- Dividing the wrong way for a unit rate.
- Using the new value as the base for percent change.
- Adding successive discount percents.

Extension Ideas

- Compare real sneaker prices online.
- Design a store logo using a scale factor.
- Track sales for a week of a school store.

Answer Key – CP2

Part 1: Best Supplier

#	Task	Answer
1	Find the price per pair for StrideCo.	$864 ÷ 24 = $36 per pair
2	Find the price per pair for RunRight and KickBox. Which supplier is cheapest?	RunRight $34.50, KickBox $37; RunRight is cheapest.
3	Socks cost $27 for 12 pairs. How much do 30 pairs cost?	$\frac{27}{12} = \frac{x}{30} \rightarrow$ x = $67.50
4	Laces cost $37.50 for 50. What is the cost per pair of laces?	$0.75

Part 2: Scale Drawing

#	Task	Answer
5	The store is 12 inches long on the plan. How long is it really?	12 × 4 = 48 ft
6	The checkout counter is $3\frac{1}{2}$ in on the plan. How long is it?	14 ft
7	A wall shelf is 20 ft long. How long should it be on the plan?	5 in
8	Draw a simple plan of the store with three labeled items and their real lengths.	Answers vary; check the 1 in = 4 ft scale.

Part 3: Prices & Raffle

#	Task	Answer
9	A pair costs the store $35. What is the price after a 60% markup?	35 + 0.60(35) = $56
10	The pair goes on sale for 25% off $56. What is the sale price?	56 − 14 = $42
11	The raffle has 200 tickets and 4 winning tickets. What is the probability of winning with 1 ticket? Write it as a fraction and a percent.	$\frac{4}{200} = \frac{1}{50}$ = 2%

Reflection

#	Prompt	Look for
R1	Why does a unit rate make comparing suppliers easier?	It shows the cost of the same amount (one pair), so prices can be compared directly.

#	Prompt	Look for
R2	Is 25% off and then 25% more the same price? Explain.	No; 25% more is taken of a smaller amount, so the final price is less than the original.

Answer Key – CP1

Part 1: Ordering

#	Task	Answer
1	Find the unit price of each shoe supplier and rank them.	RunRight $34.50 < StrideCo $36 < KickBox $37
2	Find the cost of 180 pairs from RunRight. What percent is saved compared to StrideCo?	180 × 34.50 = $6,210 vs. $6,480; save $270 ÷ 6,480 ≈ 4.2%
3	Socks: 12 pairs for $27. Use a proportion to find the cost of 180 pairs.	$\frac{27}{12} = \frac{x}{180} \rightarrow$ x = $405
4	RunRight only sells in packs of 36. How many packs do you need, and how many extra pairs do you get?	180 ÷ 36 = 5 packs exactly, 0 extra

Part 2: Store Layout

#	Task	Answer
5	Find the real length, width, and floor area of the store.	48 ft by 30 ft; 1,440 sq ft
6	A display table is similar to a 2 ft by 5 ft table and its long side is 7.5 ft. Find its short side and the scale factor.	Scale factor 1.5; short side 3 ft
7	What percent of the floor does a 6 ft by 30 ft wall of shelves cover?	180 ÷ 1,440 = 12.5%
8	If the plan scale changed to 1 in = 3 ft, how long would the store be on the plan?	48 ÷ 3 = 16 in

Part 3: Pricing & Raffle

#	Task	Answer
9	A RunRight pair costs $34.50. Find the price, then the price after a 30% grand-opening discount. Is the store still making money?	$55.20; sale $38.64 > $34.50, so yes, profit $4.14
10	Sales were 140 pairs last week and 182 pairs this week. Find the percent change.	$\frac{42}{140}$ = 30% increase
11	The raffle has 300 tickets and 5 prizes. You hold 12 tickets. What is the probability that a given drawing picks one of your tickets? What is the theoretical number of times you win if the drawing were repeated 50 times?	$\frac{12}{300} = \frac{1}{25}$ = 4%; about 2 times

Reflection

#	Prompt	Look for
R1	Explain how a markup and a discount of the same percent can still leave a profit.	They are percents of different amounts; a 60% markup then 30% off still leaves the price above cost.
R2	How do scale factors connect similar figures and scale drawings?	Both multiply every length by the same factor, keeping shapes the same.

Answer Key – Honors

Part 1: Purchasing Strategy

#	Task	Answer
1	Find the cheapest way to buy at least 200 pairs if each supplier only sells full packs. Show the cost.	Cheapest combination with at least 200 pairs: 5 RunRight packs (180 pairs, $6,210) + 1 StrideCo pack (24 pairs, $864) = 204 pairs for $7,074, which is $74 over budget. (Others: 6 RunRight = $7,452; 5 RunRight + 2 KickBox = $7,320.)
2	Using a 4% bulk discount on orders over $7,000, does the cheapest option fit the budget?	7,074 × 0.96 = $6,791.04; yes
3	Find the cost per pair for that order after the discount.	6,791.04 ÷ 204 ≈ $33.29 per pair
4	Shoe boxes come in packs of 100 for $85. How many packs are needed for 204 pairs, and what do they cost? What would 204 boxes cost at the same unit rate?	3 packs = $255. Unit rate $\frac{85}{100}$ = $0.85, so 204 boxes would cost $173.40.

Part 2: Layout & Area

#	Task	Answer
5	Find the real area and the area on the plan. What is the ratio of real area to plan area (in sq ft per sq in)?	Real 1,440 sq ft; plan 90 sq in; 16 sq ft per sq in (4^2)
6	The owner wants a second store with every length 1.25 times as long. Find its area and the percent increase in area.	1,440 × 1.5625 = 2,250 sq ft; 56.25% increase
7	A circular display has a 3 ft radius in store 1. In the larger store it is scaled by 1.25. What is the percent increase in its area?	1.25^2 = 1.5625 → 56.25%
8	Explain why area increases by more than 25% when lengths increase by 25%.	Area multiplies two lengths, so the factor is 1.25 × 1.25 = 1.5625.

Part 3: Pricing Analysis

#	Task	Answer
9	A $35 pair is marked up 60%, discounted 30%, then given an extra 10% off at checkout. Find the final price and the single equivalent discount from $56.	56 × 0.7 × 0.9 = $35.28; 1 − 0.63 = 37% off
10	The raffle: 300 tickets, 5 prizes drawn without replacement. You hold 12 tickets. Find the probability that the first two drawings are both yours.	$\frac{12}{300} \times \frac{11}{299} = \frac{132}{89{,}700}$ ≈ 0.0015 (0.15%)
11	Recommend a grand-opening price for the $34.50 pair so that the store earns at least a 20% profit on cost after a 30% discount. Show your inequality.	0.7p ≥ 1.2(34.50) = 41.40 → p ≥ $59.14; set the price at $59.99.

Reflection

#	Prompt	Look for
R1	Why are two successive discounts not the same as adding the percents?	The second percent is taken from the already-reduced price, so 30% then 10% equals 37%, not 40%.
R2	How does probability change when tickets are not replaced?	Both the number of your tickets and the total decrease, so the second probability depends on the first result.